Yash Bavadiya

+91-9879163975 krbavadiya11@gmail.com Portfolio LinkedIn GitHub X LeetCode

EDUCATION

Indian Institute of Technology, Jodhpur **Aug 2024 – Jul 2028**
Bachelor of Technology - Computer Science and Engineering - **CGPA: 7.21** Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, Rust, TypeScript, JavaScript, Python, C, SQL
Systems & AI: Distributed Systems, Discrete-Event Simulation, Concurrency, OS, Networks, DBMS, RAG, LLM Agents, MCP
Technologies: Qt, React, Node.js, WebAssembly, PostgreSQL, SQLite, Redis, WebSockets, Docker, Linux, GitHub Actions

EXPERIENCE

Workers IO (YC F26) **Sep 2026 – Present**
Developer Relations Engineer | Deterministic simulation testing Remote (San Francisco)

- Earned a founder-led hire by building **Breakscale**, an open-source system design simulator, then grew it as core maintainer to **2,000+ GitHub stars**, 168 forks and **1,200+ tests**, merging 31 PRs from **15 outside contributors**.
- Engineered a **discrete-event engine** (a time-ordered event queue) where queue buildup, timeouts, retry storms and circuit-breaker trips emerge from simulation, not formulas, across **33 component types**.
- Shipped **fault injection** (crash, slow, partition, packet loss) with **deterministic replay links**: the same seed and fault log replay the exact failure from one shareable URL.
- Launched **breakscale-mcp**, an **MCP server** that lets AI assistants model a codebase as a runnable simulation, plus a **VS Code extension**; Breakscale is in the Vercel, Claude and Mintlify open-source programs.

Google Summer of Code 2026 **May 2026 – Aug 2026**
Kdenlive @KDE Community | Open Source Contributor, C++ and Qt Remote

- Designed and shipped **per-channel RGB curves**, a **multi-stop gradient editor** and **time-remap keyframe types** in **C++ and Qt**, wired into Kdenlive's MLT filter pipeline and merged for **Kdenlive 26.08** (41,000 monthly Flathub installs).
- Resolved an **11-year-old bug** (reported 2015) with an interval-merging **gap-detection algorithm**, collaborating with core maintainers through review to merge **23 merge requests**.

Undergraduate Researcher **Jul 2025 – Nov 2025**
Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder Jodhpur

- Developed the retrieval pipeline of an **offline RAG system** over 300-page PDFs with **FAISS** vector search and a **4-bit quantized Phi-3**, answering in ∼**3 s** on a laptop GPU; benchmarked models and prompts on ROUGE (**+71.77%** over baseline).

PROJECTS

HatchZip | GitHub | Website | **Rust, Slint, ZIP/7z/RAR/tar** **Oct 2026**

- Architected an archive extractor that **frees disk space while extracting**: saves the index aside, extracts from the last entry backwards and **truncates the archive** after each batch, **punching holes** in `.tar.gz` streams.
- Guaranteed **crash safety** through ordered writes, validated by tests that kill the process at every write, save and truncate, then resume and diff every file; **130+ tests** verify freed space on ext4, APFS and NTFS.
- Achieved **5x faster** extraction than `unzip` (0.3 s vs 1.5 s on 200 MB); reached **2,700+ stars**, **27,000+ downloads** and **#1 new Rust repo** of its launch week; patched **2 security advisories**.

gyotaku | GitHub | Website | **Rust, GPUI, ONNX Runtime, SQLite FTS5** **Oct 2026**

- Created offline search over every screenshot by the text inside it, piping new screenshots through **PaddleOCR on ONNX Runtime** into a **SQLite FTS5 trigram index** that highlights matches in place.
- Optimized search to **1–10 ms per keystroke** across 5,000+ screenshots at **37 MB idle memory**; shipped on macOS, Windows and Linux with **390+ stars** and **4,000+ downloads**.

Flock | GitHub | npm | **TypeScript, WebSocket, Redis, React, Docker** **Jun 2026**

- Implemented multiplayer cursors and presence in a **10-line React integration**, scaling servers horizontally via a **Redis pub/sub relay** with **keyspace notifications** evicting dropped clients.
- Delivered throttling, mid-animation cursor smoothing and **exponential-backoff reconnects**; published 3 npm packages with **1,100+ downloads** and **58 tests**.

Game Boy Emulator | GitHub | **C++20, SDL2, CMake** **Jul 2026**

- Emulated the Game Boy from scratch: a CPU with all **501 opcodes**, 5 interrupts, a DIV-based timer, PPU and **MBC1/MBC3** banking; passes all **11 Blargg CPU test ROMs** and **dmg-acid2**, and boots Pokémon Red.

ACHIEVEMENTS

- **Open source:** contributed fixes merged into Deno, Adobe React Spectrum, Kubernetes SIGs, nlohmann/json, Vitest, Nuxt and jj.
- **Building in public:** grew to **3,000+ followers** on X and **3.9M impressions** in 3 months; HatchZip's launch hit **1.9M views**.
- Ranked AIR **3449** of 180,000+ in JEE Advanced 2024; JEE Main **99.65** percentile; **450+** LeetCode solved; Codeforces max **1429**.
- Mentored **30 students** into open source and led 2 workshops at DevlUp Labs; Core Member, Programming Club and E-Cell.